

BUSINESS REQUIREMENTS SPECIFICATION (BRS)

Project: Virtual Worker Platform

Technologies: n8n + Hugging Face

1. EXECUTIVE SUMMARY

1.1 Purpose

The purpose of this project is to design and implement a governed, scalable Virtual Worker platform capable of completing structured business and operational tasks on behalf of users and companies.

The system will: - Accept natural language task requests - Interpret intent using AI (Hugging Face models) - Map requests to controlled workflow templates - Execute workflows via n8n - Enforce approvals and governance controls - Provide full auditability and traceability

2. BUSINESS OBJECTIVES

1. Reduce manual operational workload by at least 50% for repetitive digital tasks.
 2. Enable organizations to deploy automation without writing custom code.
 3. Provide enterprise-grade governance and approval controls over AI-driven actions.
 4. Create a reusable template-based execution engine that scales across departments.
 5. Establish a foundation for a long-term AI-powered workforce platform.
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3. PROJECT SCOPE

3.1 In Scope (MVP)

- User and workspace management (multi-tenant)
- Role-Based Access Control (RBAC)
- Task intake (UI + API)
- AI-powered classification and entity extraction
- Template-based workflow matching
- n8n workflow execution bridge
- Approval system for sensitive actions
- Audit logs and task history
- Notifications (Email/Slack)

- Basic monitoring and logging

3.2 Out of Scope (Phase 1)

- Fully autonomous AI with unrestricted tool execution
 - Robotic desktop automation (RPA)
 - Complex multi-agent orchestration
 - Financial transaction processing without approval gates
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4. STAKEHOLDERS

- Product Owner
 - Technical Lead
 - Automation Engineer (n8n)
 - AI Engineer (HF integration)
 - Security & Compliance Lead
 - End Users
 - Approvers/Managers
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5. USER ROLES

5.1 Admin

- Manage workspace
- Create/edit templates
- Manage users
- View audit logs

5.2 Member

- Create tasks
- View own task history

5.3 Approver

- Approve or reject sensitive task executions

5.4 Viewer

- Read-only access to task results
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6. FUNCTIONAL REQUIREMENTS

FR-01: User Authentication & Workspace Isolation

- Secure login
- Users belong to a workspace
- All queries scoped by workspace_id

FR-02: Task Submission

Users must be able to: - Create task (title, description, attachments) - Set priority - View status updates

FR-03: AI Interpretation

System must: - Classify task type using HF model - Extract structured entities - Suggest template match - Request clarification if confidence below threshold

FR-04: Template Management

Admins must: - Create templates - Define required inputs - Link template to n8n workflow ID - Set approval rules - Version templates

FR-05: Workflow Execution

System must: - Trigger n8n workflow via webhook/API - Pass structured inputs - Track execution ID - Capture outputs

FR-06: Approval Workflow

System must: - Pause workflow if approval required - Notify approver - Resume execution on approval - Log decision

FR-07: Audit Trail

System must log: - Task creation - AI classification result - Template match - Workflow start - Approval events - Completion or failure

FR-08: Notifications

System must notify: - Approval requests - Task completion - Task failure

7. NON-FUNCTIONAL REQUIREMENTS

7.1 Security

- Encrypted credentials
- Multi-tenant isolation
- Approval enforcement
- No direct AI tool execution

7.2 Performance

- AI interpretation < 5 seconds
- Workflow response time tracked

7.3 Reliability

- Retry logic in n8n
- Failure alerts
- Execution timeouts

7.4 Scalability

- n8n queue mode
- Horizontal scaling capability
- Modular architecture

7.5 Observability

- Centralized logging
 - Metrics dashboard
 - Error tracking
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8. SUCCESS CRITERIA

Technical

- 95% workflow success rate
- <5s AI interpretation
- 0 cross-tenant breaches

Business

- 50% reduction in manual processing
 - 70% automation rate without manual correction
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END OF BRS